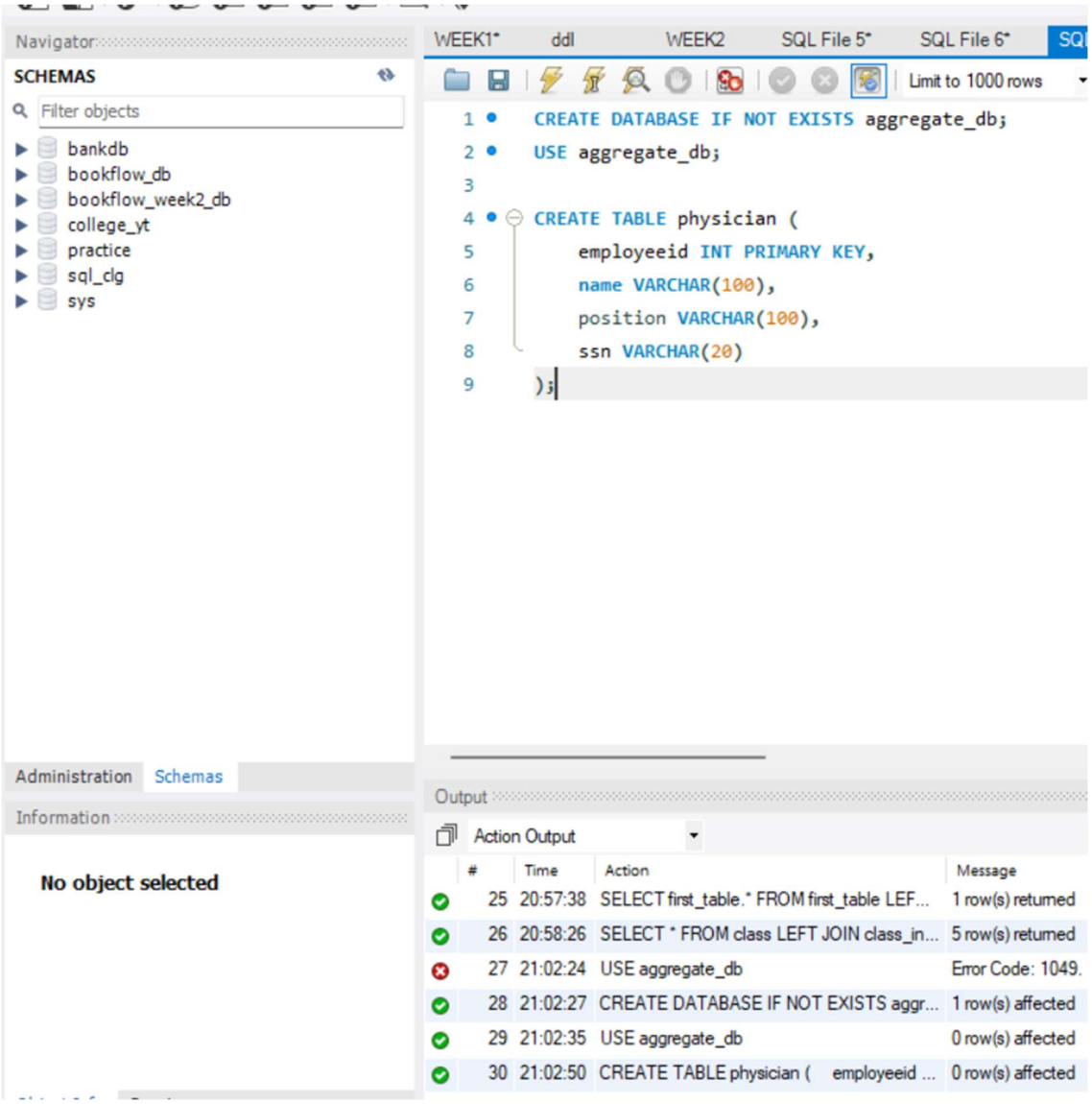
SQL PRACTICE NOTES - PHYSICIAN & DEPARTMENT TABLES

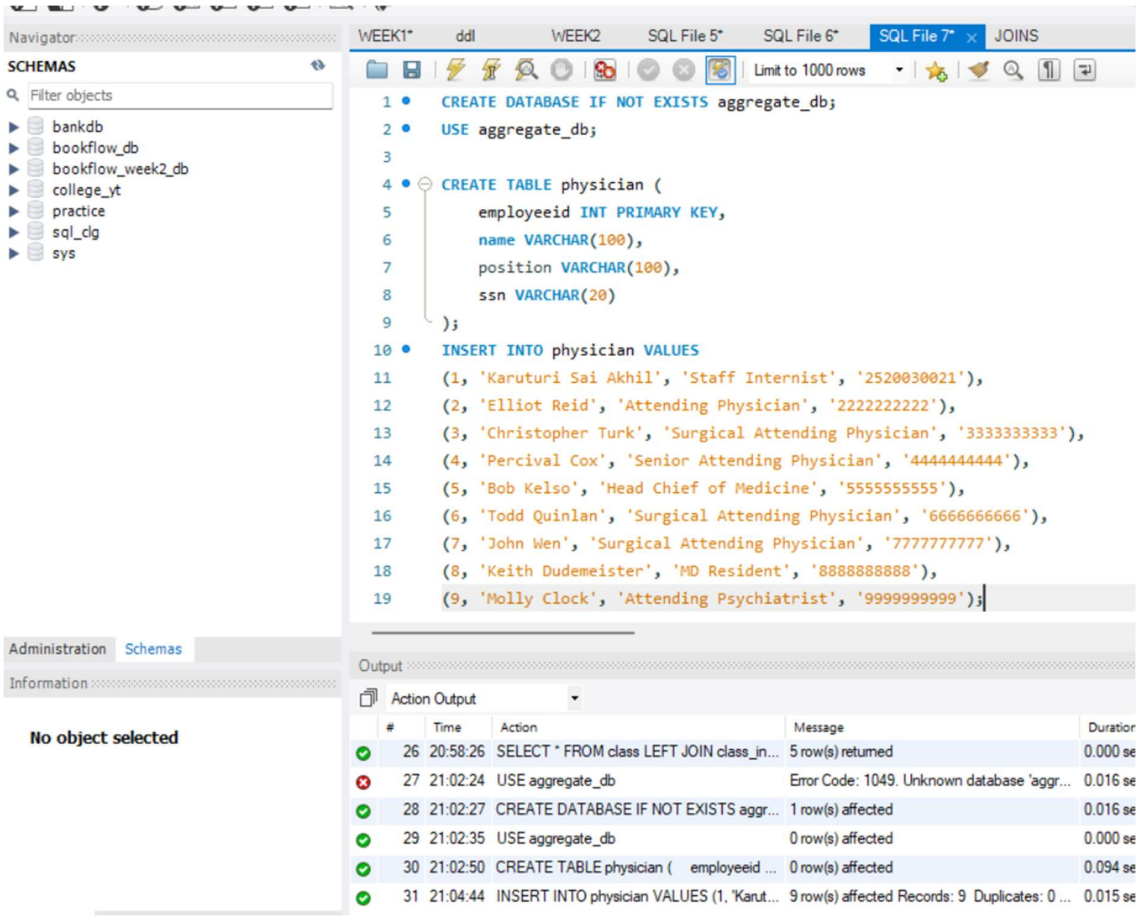
2520030071

G. Hemanth Venkatesh

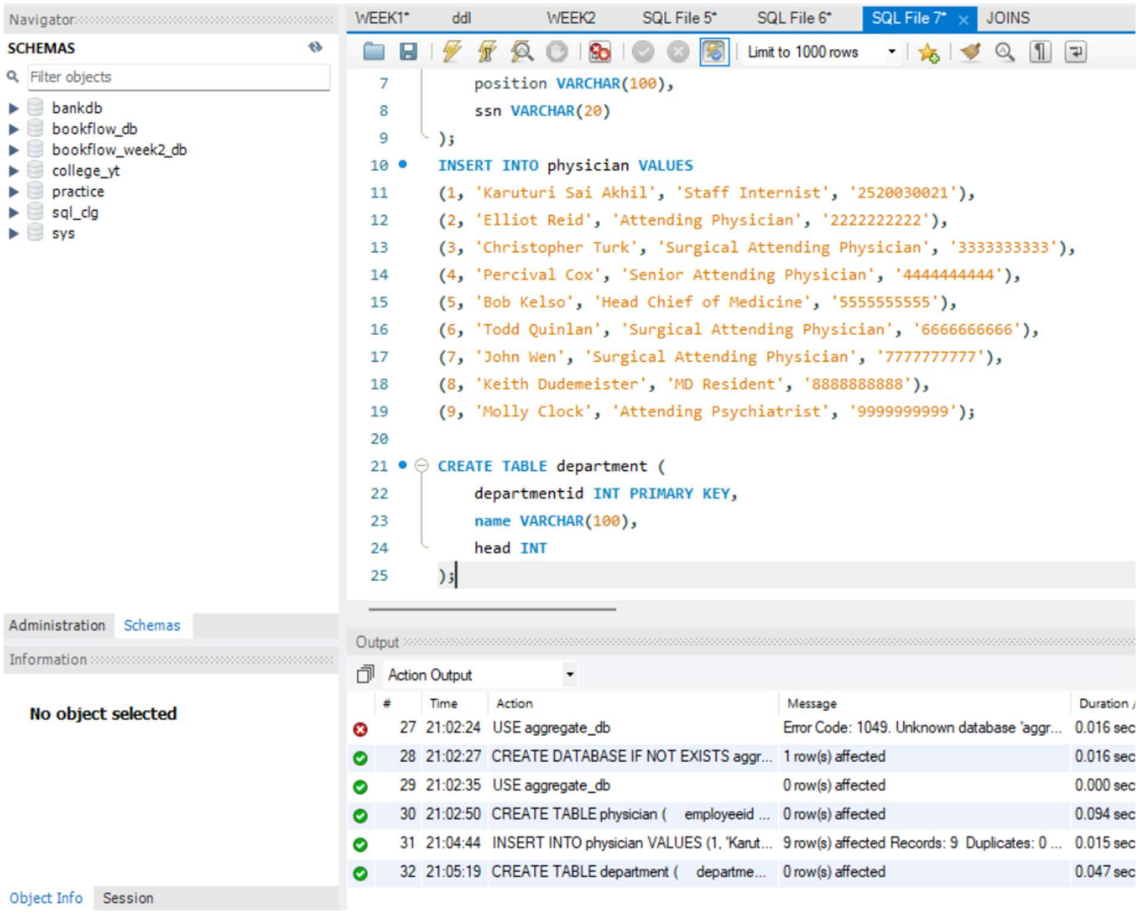
STEP 1: CREATE PHYSICIAN TABLE



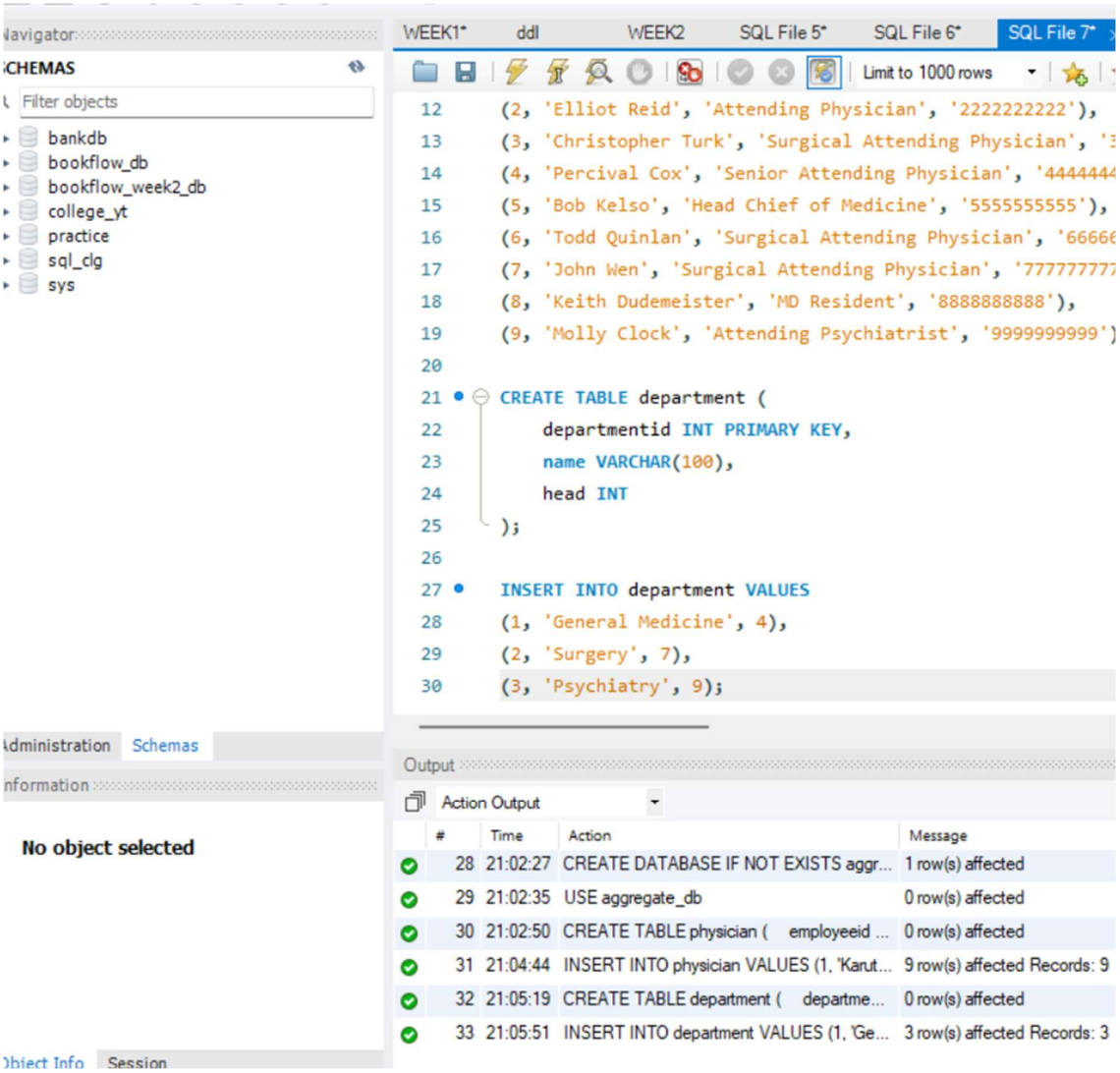
STEP 2: INSERT DATA INTO PHYSICIAN



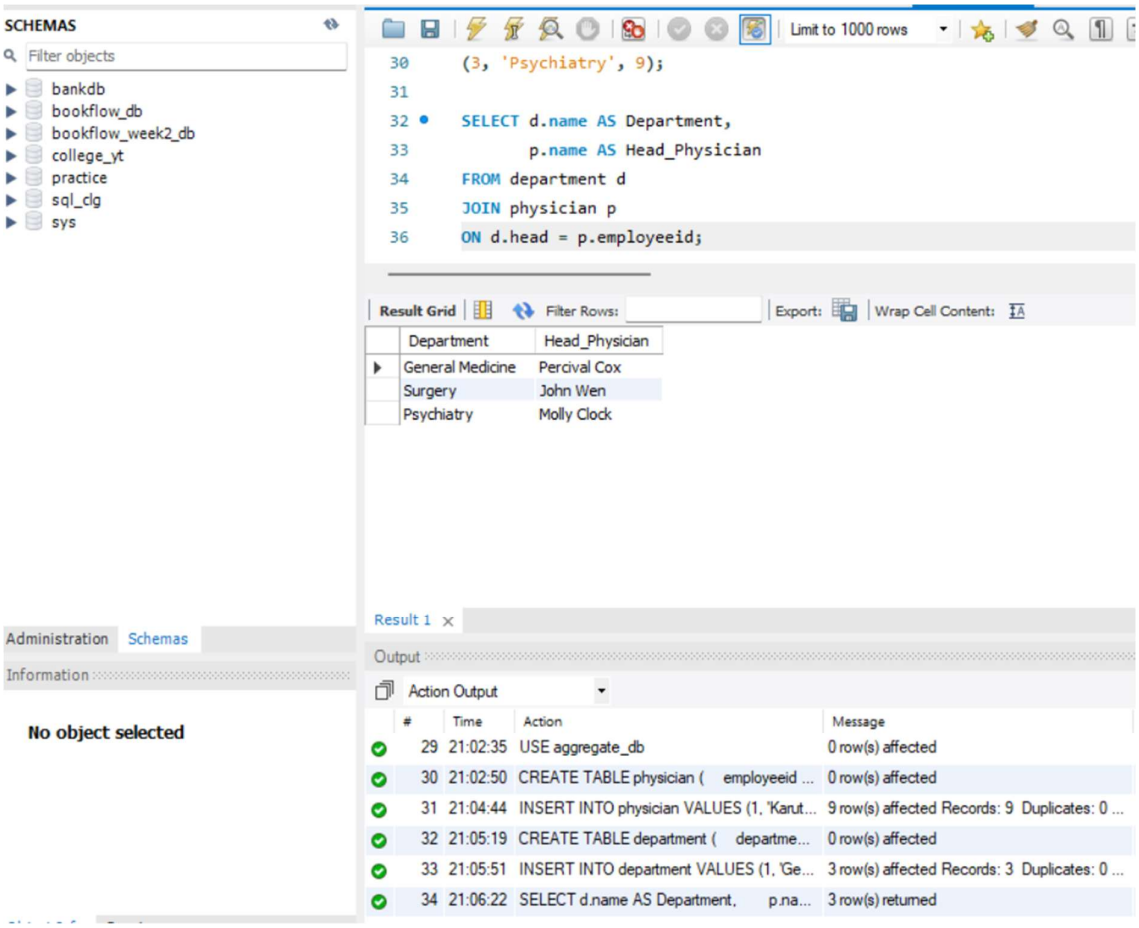
STEP 3: CREATE DEPARTMENT TABLE



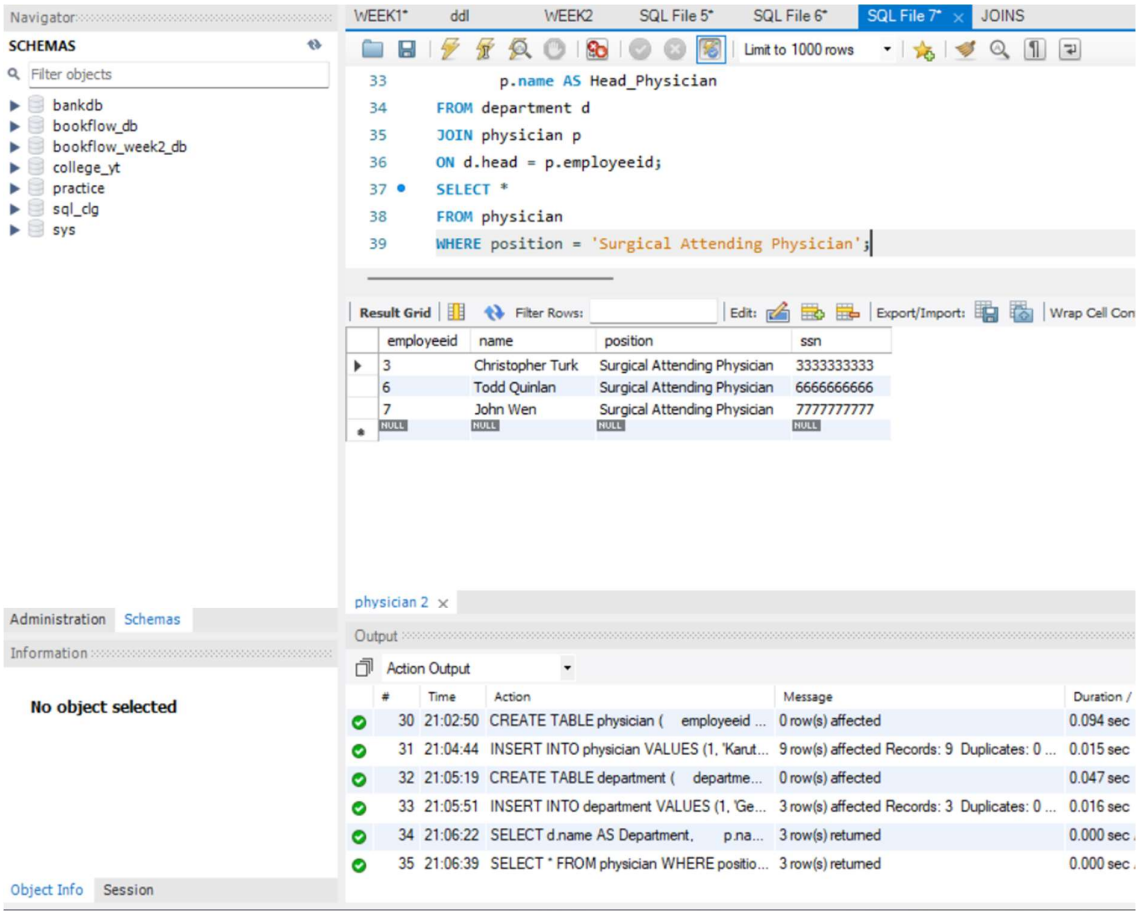
STEP 4: Insert Department Data


WEEK1*
ddl
WEEK2
SQL File 5*
SQL File 6*
SQL File 7*
Limit to 1000 rows
(2, 'Elliot Reid', 'Attending Physician', '2222222222'),
(3, 'Christopher Turk', 'Surgical Attending Physician',
(4, 'Percival Cox', 'Senior Attending Physician',
(5, 'Bob Kelso', 'Head Chief of Medicine', '5555555555'),
(6, 'Todd Quinlan', 'Surgical Attending Physician',
(7, 'John Wen', 'Surgical Attending Physician',
(8, 'Keith Dudemeister', 'MD Resident', '8888888888'),
(9, 'Molly Clock', 'Attending Psychiatrist', '9999999999')
CREATE TABLE department (
departmentid INT PRIMARY KEY,
name VARCHAR(100),
head INT
);
INSERT INTO department VALUES
(1, 'General Medicine', 4),
(2, 'Surgery', 7),
(3, 'Psychiatry', 9);
CHEMAS
Filter objects
bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys
dministration
Schemas
nformation
No object selected
Object Info
Session
Output
Action Output
Time Action Message
28 21:02:27 CREATE DATABASE IF NOT EXISTS aggr... 1 row(s) affected
29 21:02:35 USE aggregate_db 0 row(s) affected
30 21:02:50 CREATE TABLE physician (employeeid ... 0 row(s) affected
31 21:04:44 INSERT INTO physician VALUES (1, 'Karut... 9 row(s) affected Records: 9
32 21:05:19 CREATE TABLE department (departme... 0 row(s) affected
33 21:05:51 INSERT INTO department VALUES (1, 'Ge... 3 row(s) affected Records: 3

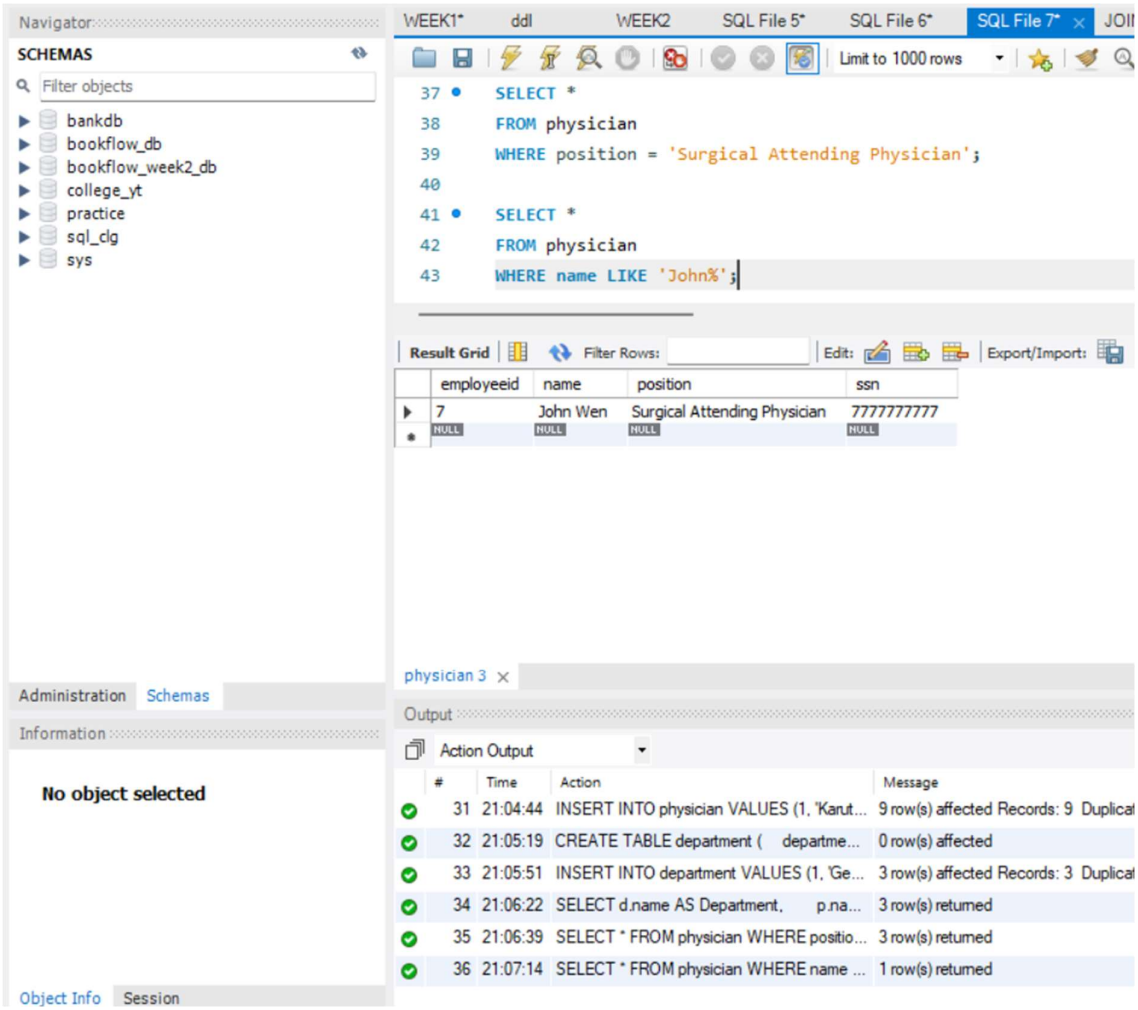
1. INNER JOIN



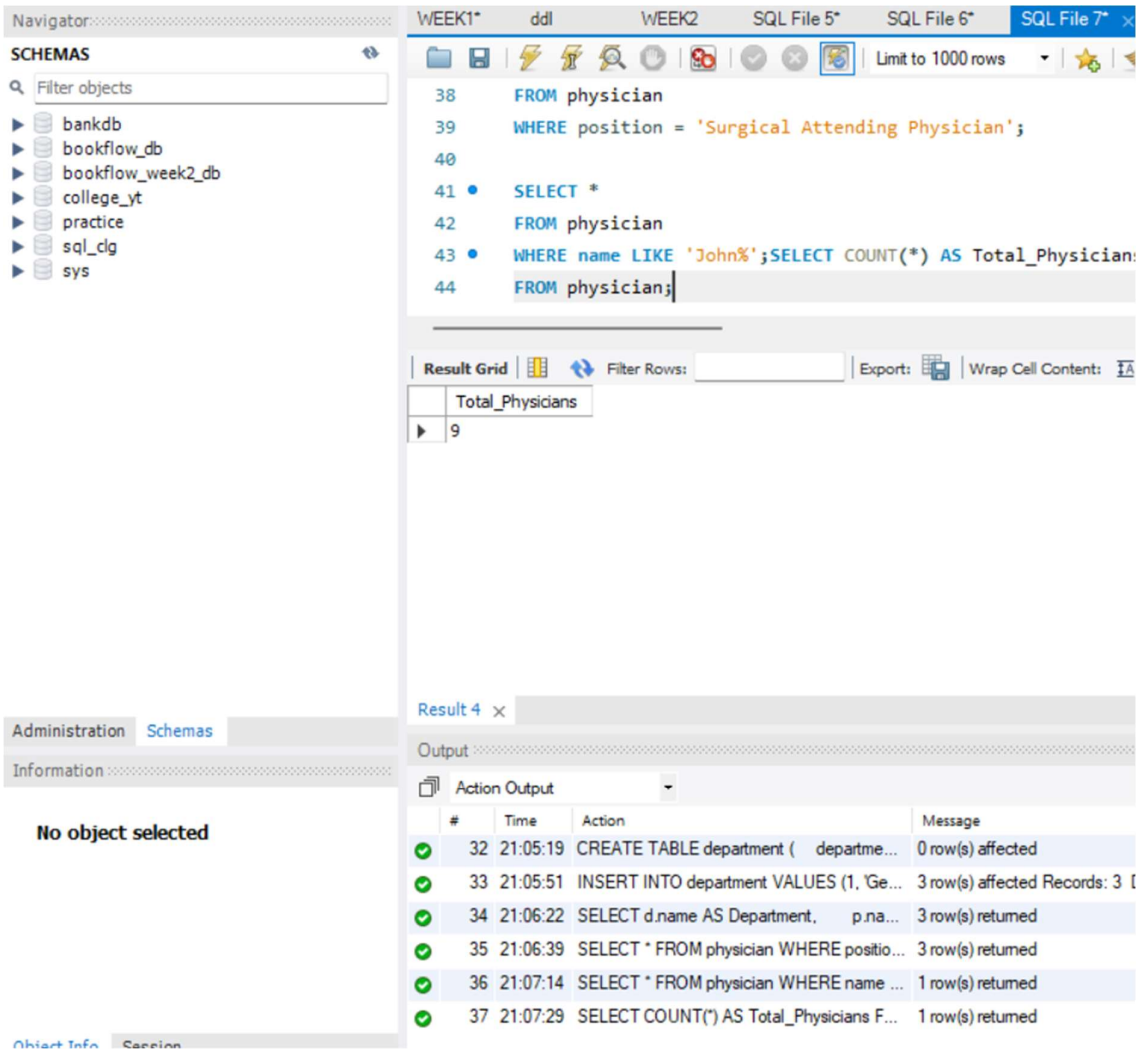
2. WHERE CLAUSE


Navigator
SCHEMAS
Filter objects
bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys
WEEK1*
ddl
WEEK2
SQL File 5*
SQL File 6*
SQL File 7*
JOINS
Limit to 1000 rows
33 p.name AS Head_Physician
34 FROM department d
35 JOIN physician p
36 ON d.head = p.employeeid;
37 SELECT *
38 FROM physician
39 WHERE position = 'Surgical Attending Physician';
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Con
employeeid name position ssn
3 Christopher Turk Surgical Attending Physician 3333333333
6 Todd Quinlan Surgical Attending Physician 6666666666
7 John Wen Surgical Attending Physician 7777777777
NULL NULL NULL NULL
physician 2
Administration
Schemas
Information
No object selected
Object Info
Session
Output
Action Output
Time Action Message Duration /
30 21:02:50 CREATE TABLE physician (employeeid ... 0 row(s) affected 0.094 sec
31 21:04:44 INSERT INTO physician VALUES (1, 'Karut... 9 row(s) affected Records: 9 Duplicates: 0 ... 0.015 sec
32 21:05:19 CREATE TABLE department (departme... 0 row(s) affected 0.047 sec
33 21:05:51 INSERT INTO department VALUES (1, 'Ge... 3 row(s) affected Records: 3 Duplicates: 0 ... 0.016 sec
34 21:06:22 SELECT d.name AS Department, p.na... 3 row(s) returned 0.000 sec
35 21:06:39 SELECT * FROM physician WHERE positio... 3 row(s) returned 0.000 sec

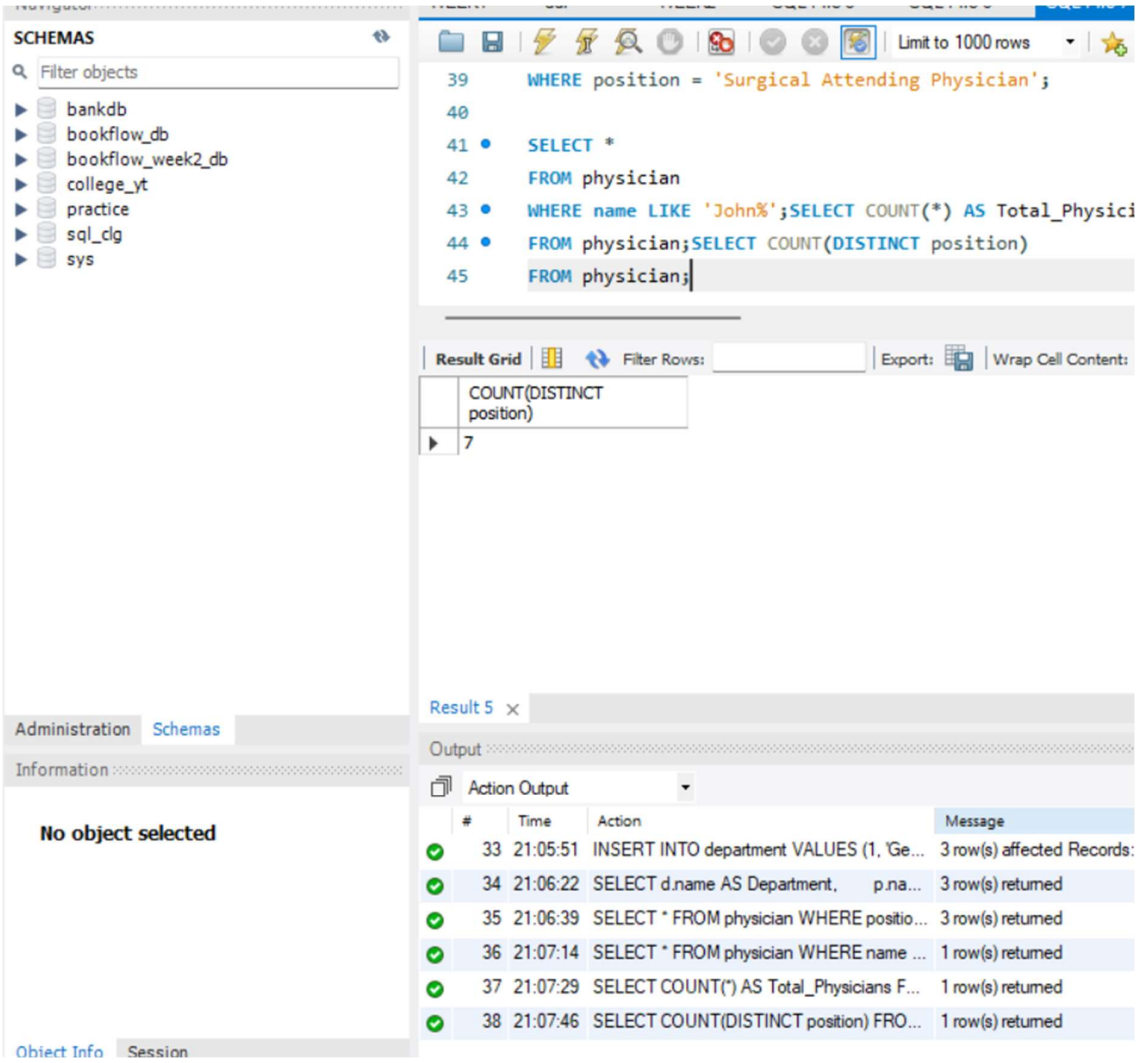
WHERE + LIKE


Navigator:
SCHEMAS
Filter objects
bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys
Administration
Schemas
Information
No object selected
Object Info
Session
WEEK1*
ddl
WEEK2
SQL File 5*
SQL File 6*
SQL File 7*
JOII
Limit to 1000 rows
37 SELECT *
38 FROM physician
39 WHERE position = 'Surgical Attending Physician';
40
41 SELECT *
42 FROM physician
43 WHERE name LIKE 'John%';
Result Grid
Filter Rows:
Edit:
Export/Import:
employeeid
name
position
ssn
7
John Wen
Surgical Attending Physician
7777777777
NULL
physician 3
Output
Action Output
Time Action Message
31 21:04:44 INSERT INTO physician VALUES (1, 'Karut... 9 row(s) affected Records: 9 Duplicat
32 21:05:19 CREATE TABLE department (departme... 0 row(s) affected
33 21:05:51 INSERT INTO department VALUES (1, 'Ge... 3 row(s) affected Records: 3 Duplicat
34 21:06:22 SELECT d.name AS Department, p.na... 3 row(s) returned
35 21:06:39 SELECT * FROM physician WHERE positio... 3 row(s) returned
36 21:07:14 SELECT * FROM physician WHERE name ... 1 row(s) returned

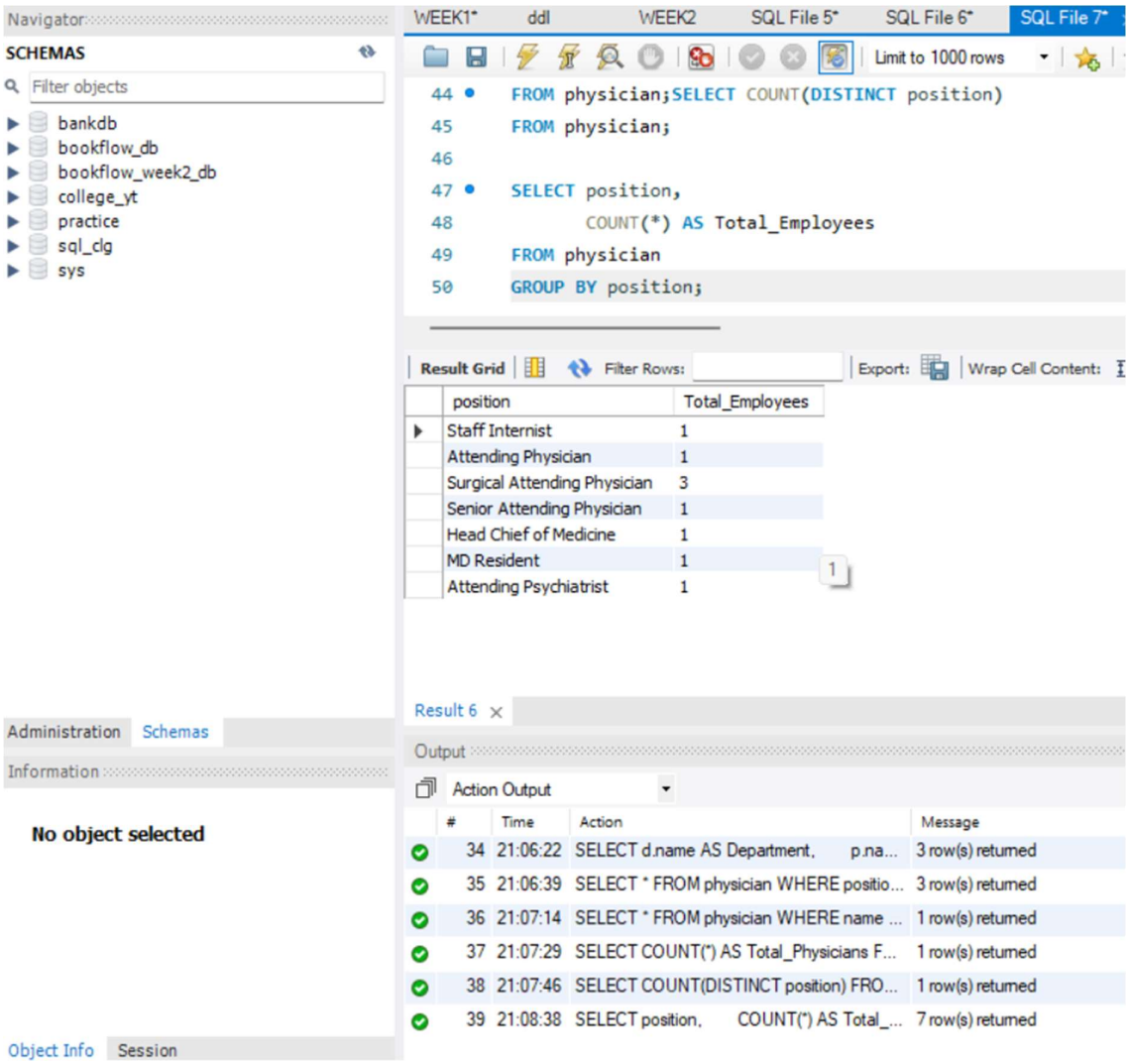
COUNT()



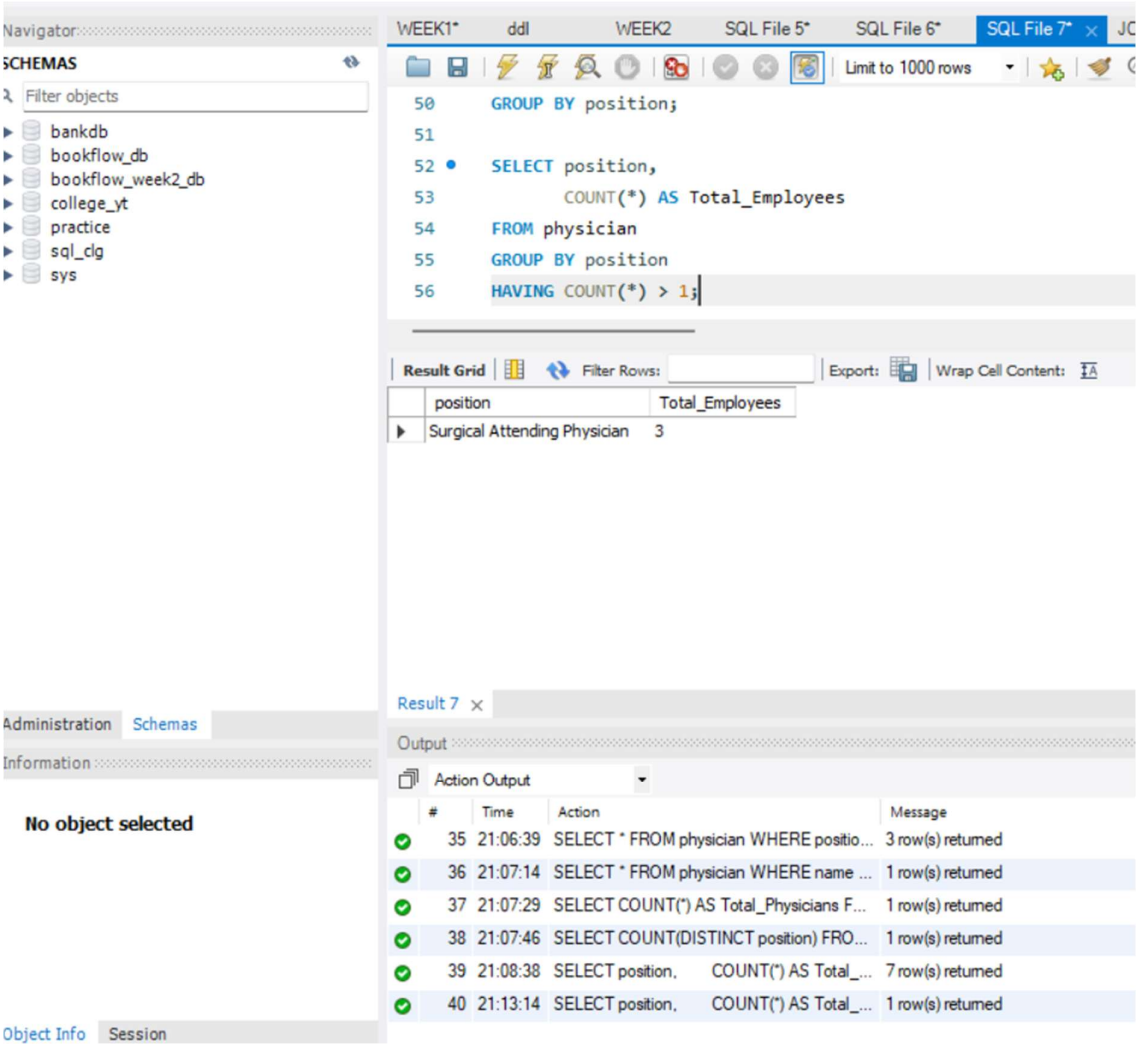
COUNT DISTINCT


SCHEMAS
Filter objects
bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys
Administration
Schemas
Information
No object selected
Object Info
Session
Limit to 1000 rows
39 WHERE position = 'Surgical Attending Physician';
40
41 SELECT *
42 FROM physician
43 WHERE name LIKE 'John%';SELECT COUNT(*) AS Total_Physici
44 FROM physician;SELECT COUNT(DISTINCT position)
45 FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
COUNT(DISTINCT position)
7
Result 5
Output
Action Output
Time Action Message
33 21:05:51 INSERT INTO department VALUES (1, 'Ge... 3 row(s) affected Records:
34 21:06:22 SELECT d.name AS Department, p.na... 3 row(s) returned
35 21:06:39 SELECT * FROM physician WHERE positio... 3 row(s) returned
36 21:07:14 SELECT * FROM physician WHERE name ... 1 row(s) returned
37 21:07:29 SELECT COUNT(*) AS Total_Physicians F... 1 row(s) returned
38 21:07:46 SELECT COUNT(DISTINCT position) FRO... 1 row(s) returned

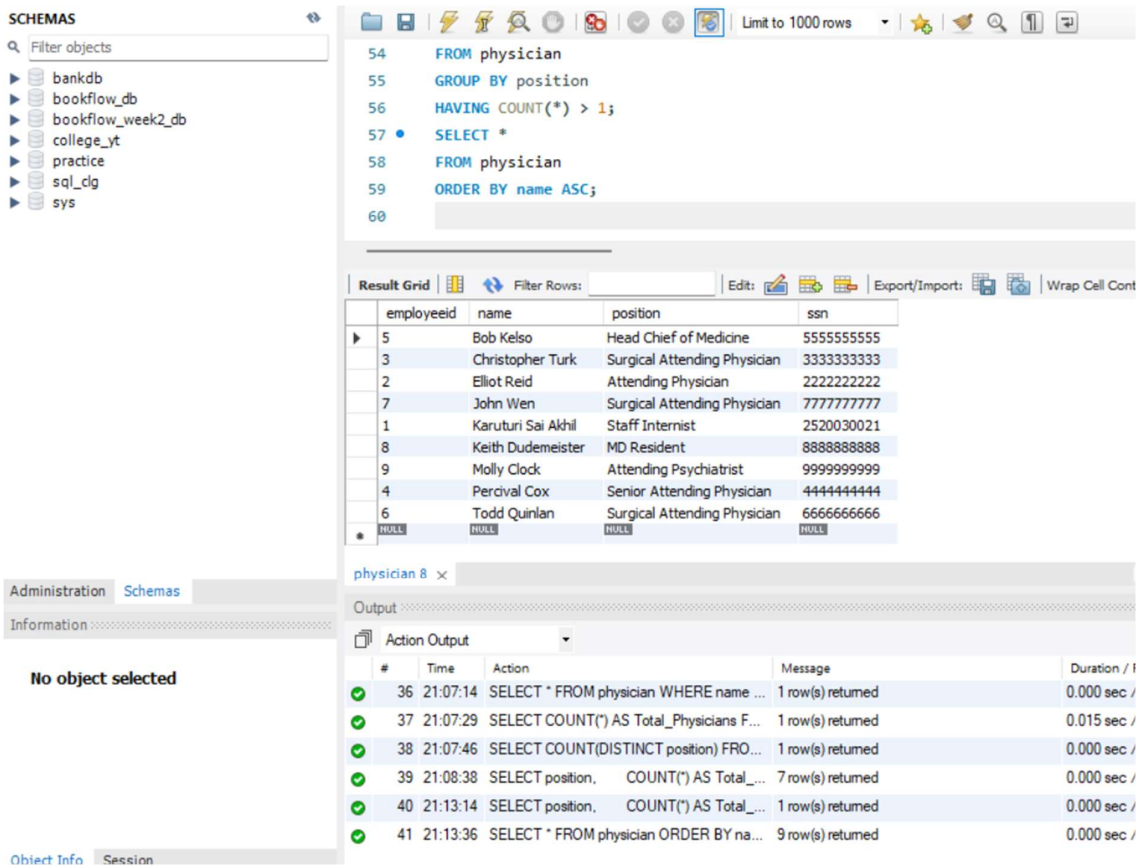
Group By



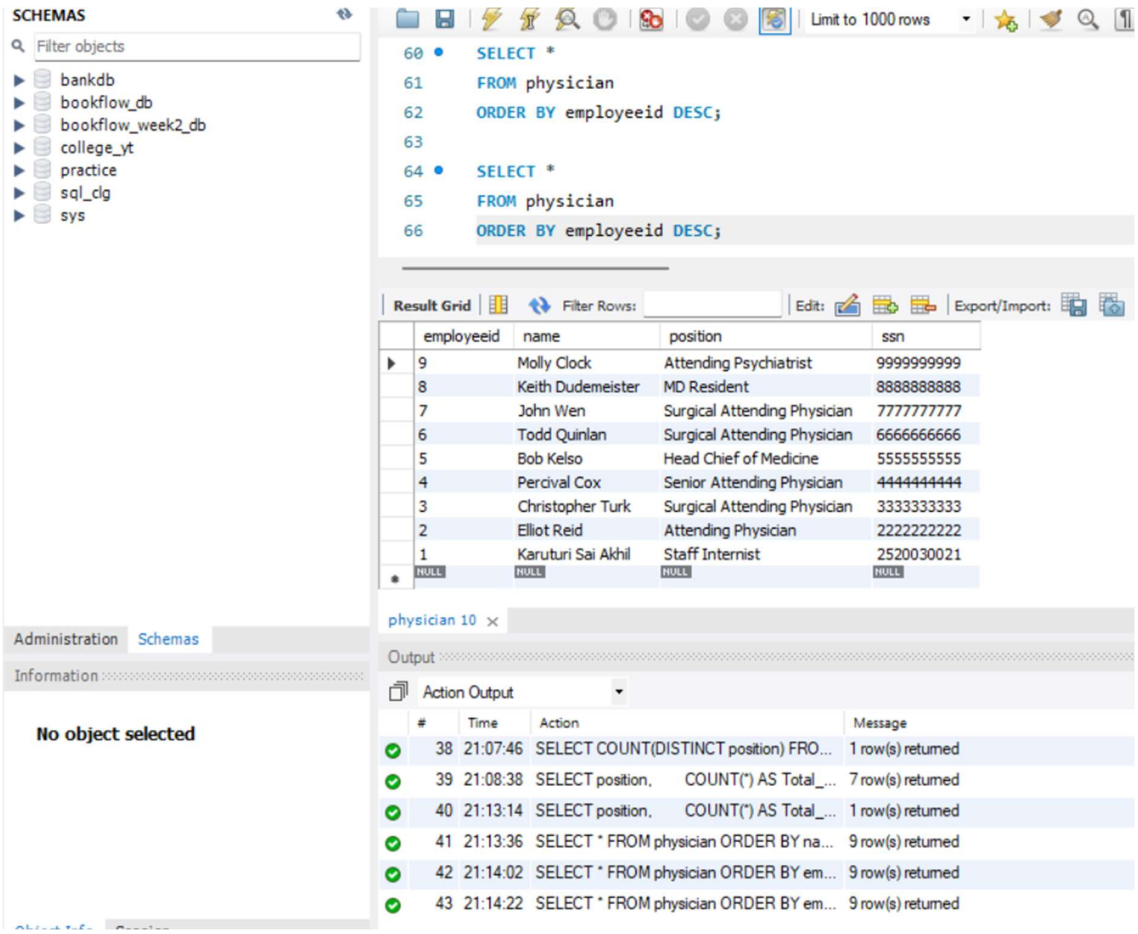
Group By + Having


Navigator
SCHEMAS
Filter objects
bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys
Administration
Schemas
Information
No object selected
Object Info
Session
WEEK1*
ddl
WEEK2
SQL File 5*
SQL File 6*
SQL File 7*
Limit to 1000 rows
50 GROUP BY position;
51
52 SELECT position,
53 COUNT(*) AS Total_Employees
54 FROM physician
55 GROUP BY position
56 HAVING COUNT(*) > 1;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
position
Total_Employees
Surgical Attending Physician
3
Result 7
Output
Action Output
Time Action Message
35 21:06:39 SELECT * FROM physician WHERE positio... 3 row(s) returned
36 21:07:14 SELECT * FROM physician WHERE name ... 1 row(s) returned
37 21:07:29 SELECT COUNT(*) AS Total_Physicians F... 1 row(s) returned
38 21:07:46 SELECT COUNT(DISTINCT position) FRO... 1 row(s) returned
39 21:08:38 SELECT position, COUNT(*) AS Total_... 7 row(s) returned
40 21:13:14 SELECT position, COUNT(*) AS Total_... 1 row(s) returned

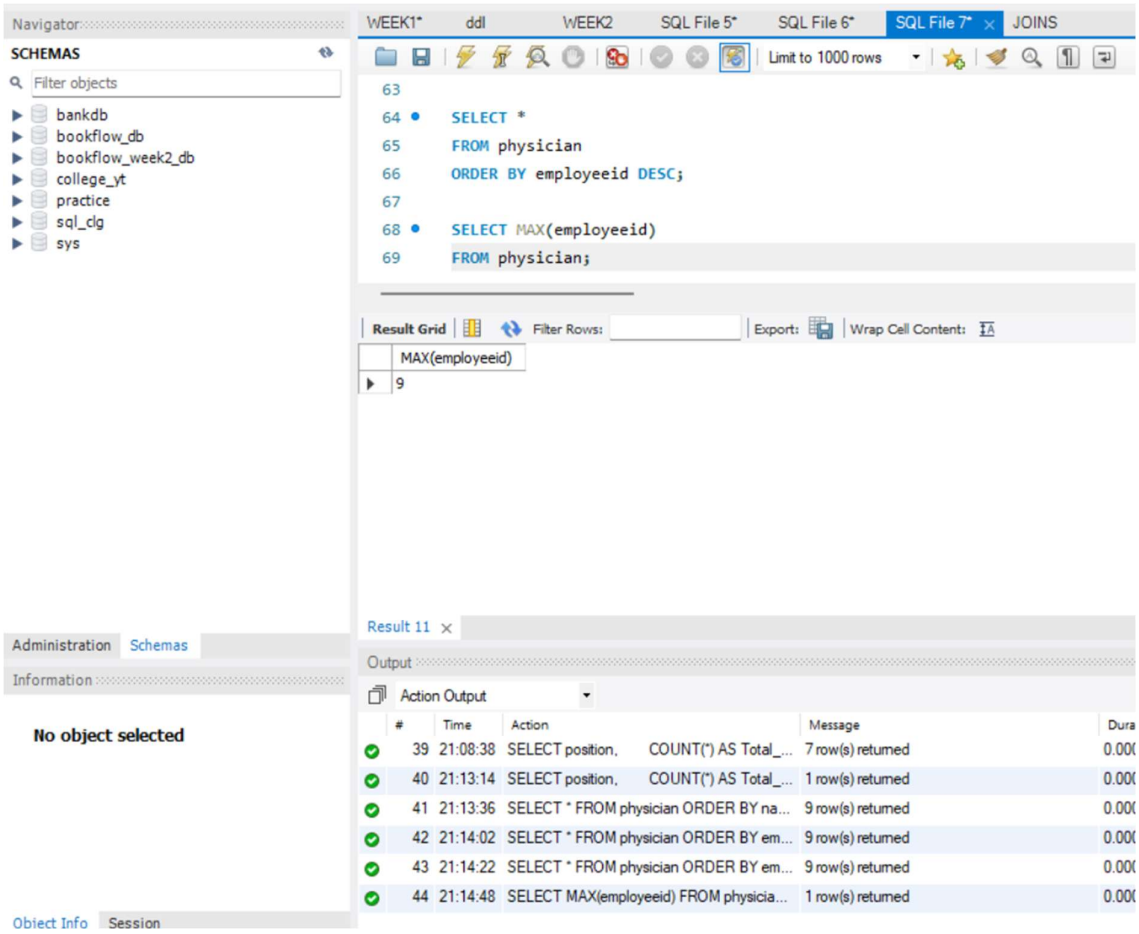
ORDER BY



ORDER BY DESC


SCHEMAS
Filter objects
bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys
Administration
Schemas
Information
No object selected
Limit to 1000 rows
60 SELECT *
61 FROM physician
62 ORDER BY employeeid DESC;
63
64 SELECT *
65 FROM physician
66 ORDER BY employeeid DESC;
Result Grid
Filter Rows:
Edit:
Export/Import:
employeeid | name | position | ssn
9 | Molly Clock | Attending Psychiatrist | 9999999999
8 | Keith Dudemeister | MD Resident | 8888888888
7 | John Wen | Surgical Attending Physician | 7777777777
6 | Todd Quinlan | Surgical Attending Physician | 6666666666
5 | Bob Kelso | Head Chief of Medicine | 5555555555
4 | Percival Cox | Senior Attending Physician | 4444444444
3 | Christopher Turk | Surgical Attending Physician | 3333333333
2 | Elliot Reid | Attending Physician | 2222222222
1 | Karuturi Sai Akhil | Staff Internist | 2520030021
physician 10
Output
Action Output
| Time | Action | Message
38 21:07:46 SELECT COUNT(DISTINCT position) FRO... 1 row(s) returned
39 21:08:38 SELECT position, COUNT(*) AS Total_... 7 row(s) returned
40 21:13:14 SELECT position, COUNT(*) AS Total_... 1 row(s) returned
41 21:13:36 SELECT * FROM physician ORDER BY na... 9 row(s) returned
42 21:14:02 SELECT * FROM physician ORDER BY em... 9 row(s) returned
43 21:14:22 SELECT * FROM physician ORDER BY em... 9 row(s) returned

MAX()


Navigator:
SCHEMAS
Filter objects
bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys
Administration
Schemas
Information
No object selected
Object Info
Session
WEEK1*
ddl
WEEK2
SQL File 5*
SQL File 6*
SQL File 7*
JOINS
Limit to 1000 rows
63
64 SELECT *
65 FROM physician
66 ORDER BY employeeid DESC;
67
68 SELECT MAX(employeeid)
69 FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
MAX(employeeid)
9
Result 11
Output
Action Output
Time Action Message Dura
39 21:08:38 SELECT position, COUNT(*) AS Total_... 7 row(s) returned 0.00(
40 21:13:14 SELECT position, COUNT(*) AS Total_... 1 row(s) returned 0.00(
41 21:13:36 SELECT * FROM physician ORDER BY na... 9 row(s) returned 0.00(
42 21:14:02 SELECT * FROM physician ORDER BY em... 9 row(s) returned 0.00(
43 21:14:22 SELECT * FROM physician ORDER BY em... 9 row(s) returned 0.00(
44 21:14:48 SELECT MAX(employeeid) FROM physicia... 1 row(s) returned 0.00(